

Parking avoiding permutations with fixed lucky sets

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Joint work with Melanie Ferreri (William & Mary)

Background

Permutations

A **permutation** of $[n] = \{1, 2, \dots, n\}$ is a bijection from $[n]$ to itself.

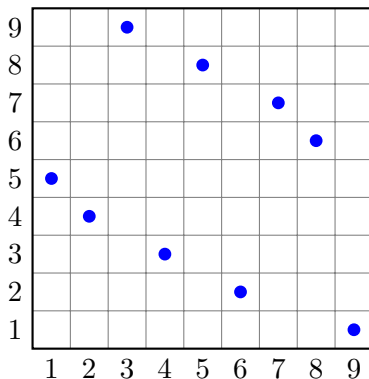


Figure: E.g. $\pi = 549382761$

Patterns

A permutation σ of size k is a pattern of a permutation π of size n if there exist $1 \leq i_1 < i_2 < \cdots < i_k \leq n$ such that $\pi_{i_1} \pi_{i_2} \cdots \pi_{i_k}$ is in the same relative order ($\equiv$) as σ .

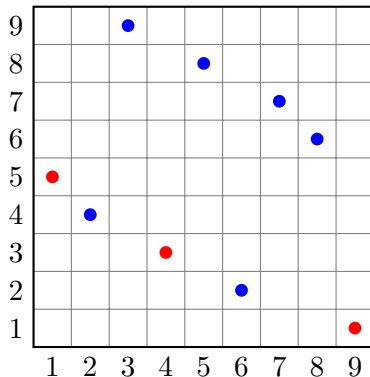


Figure: E.g. **321** is a pattern of $\pi = 549382761$ since $531 \equiv 321$

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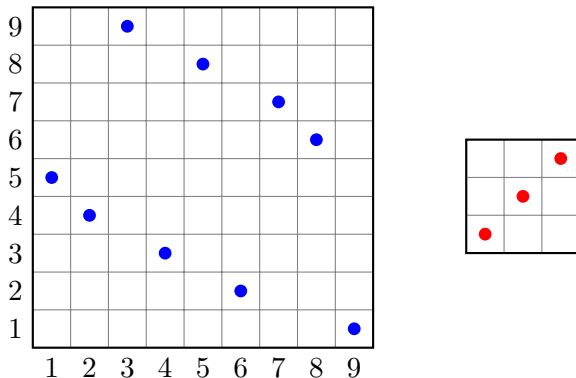


Figure: $\pi = 549382761$ avoids the pattern 123

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Result (Knuth)

For any $\sigma \in \mathfrak{S}_3$,

$$|\text{Av}_n(\sigma)| = \frac{1}{n+1} \binom{2n}{n} = C_n,$$

where C_n is the n th Catalan number.

The first terms of the sequence are:

(OEIS: [A000108](#)) 1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796, ...

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to name a few... (over 300 possible statistics listed in FindStat)

Statistics: Descents and ascents (and more)

Results

- ① Eulerian numbers (OEIS [A008292](#)): the number of permutations in $\mathfrak{S}_n$ with k descents

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Distributions of Statistics over Pattern-Avoiding Permutations

Michael Bukata Ryan Kulwicki Nicholas Lewandowski
Lara Pudwell Jacob Roth Teresa Wheeland

Valparaiso University
July 22, 2019

Abstract

We consider the distribution of ascents, descents, peaks, valleys, double ascents, and double descents over permutations avoiding a set of patterns. Many of these statistics have already been studied over sets of permutations avoiding a single pattern of length 3. However, the distribution of peaks over 321-avoiding permutations is new and we relate it statistics on Dyck paths. We also obtain new interpretations of a number of well-known combinatorial sequences by studying these statistics over permutations avoiding two patterns of length 3.

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Indeed, for any $\sigma \in \mathfrak{S}_n$, 1 cannot belong to a peak set since there is no element to the left of σ_1 . Similarly for n .

i and $i + 1$ cannot both be in the peak set since $\sigma_i > \sigma_{i+1}$ and $\sigma_i < \sigma_{i+1}$ cannot both be true.

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Theorem (Billey-Burdzy-Sagan, 2013)

The number of $\sigma \in \mathfrak{S}_n$ with peak set S is a power of two times a polynomial in n (called the peak polynomial).

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- ③ (González, Harris, Rojas Kirby, Smit Vega Garcia, and Tenner, 2023): Studied signed permutations and considered admissible pinnacle sets.

Parking Functions

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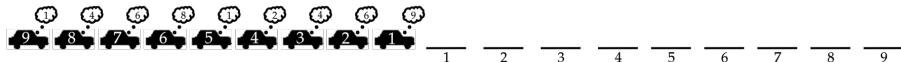
If the all the cars are able to park (under the parking rule) then $a = (a_1, a_2, \dots, a_n)$ is called a parking function.

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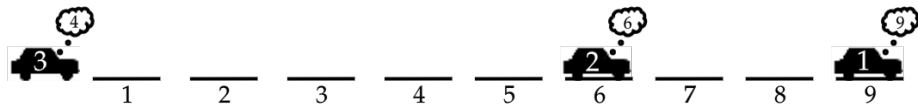
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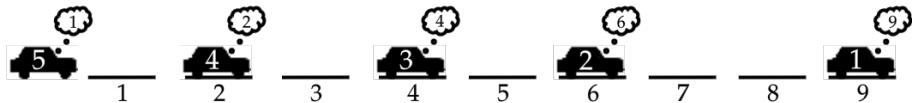
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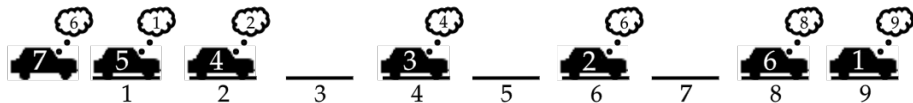
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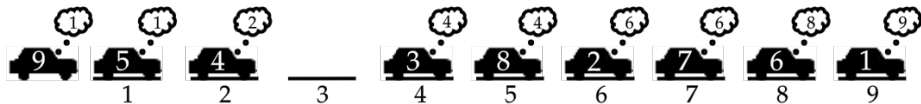
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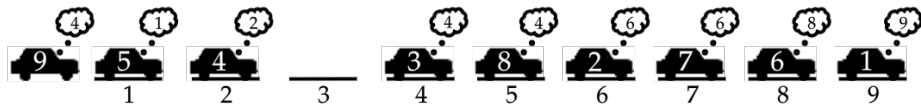
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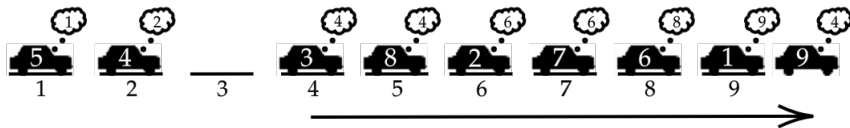
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Is $\alpha = (9, 6, 4, 2, 1, 8, 6, 4, \textcolor{red}{4})$ a parking function?



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Theorem (Pyke (1959), Konheim and Weiss (1966))

$$|PF(n)| = (n + 1)^{(n-1)}.$$

Parking permutations

Definition

If $\mathfrak{S}_n$ denotes the set of permutations of $[n]$ written in one-line notation, then the parking permutation of α is

$$\mathcal{O}(\alpha) = \pi_1 \pi_2 \cdots \pi_n,$$

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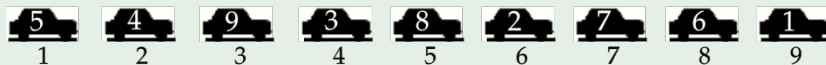
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Example

From our previous example: If $\alpha = (9, 6, 4, 2, 1, 8, 6, 4, 1)$ then the parking permutation is $\pi = 549382761$:



Pattern avoidance in parking permutations

Let $\alpha = (9, 6, 4, 2, 1, 8, 6, 4, 1)$, the following are two types of permutations arising from parking functions:

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- (M. and Zeilberger, 2026) Enumerated parking functions whose parking permutations avoid 1234 and 2134.

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- If $\alpha = (9, 6, 4, 2, 1, 8, 6, 4, 1)$ then $\pi = \boxed{5} \boxed{4} 9 \boxed{3} 8 \boxed{2} 7 \boxed{6} \boxed{1}$

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We denote the set of lucky cars by of parking functions by $I = \text{Lucky}(\alpha)$.

Lucky generating function

Theorem (Gessel and Seo, 2005)

The generating function for the number of parking functions based on the number of lucky cars is given by

$$L_n(q) = \sum_{\alpha \in PF_n} q^{\text{lucky}(\alpha)} = q \prod_{i=1}^{n-1} (i + (n - i + 1)q).$$

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This does not tell us which cars are lucky!

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For any subset $I \subseteq [n]$, we define

$$\mathcal{O}_n(I) := \{\pi \in \mathfrak{S}_n : \mathcal{O}(\alpha) = \pi \text{ for some } \alpha \in PF_n \text{ with } \text{Lucky}(\alpha) = I\},$$

i.e. $\mathcal{O}_n(I)$ denotes the set of parking permutations of length n for parking functions with lucky set I .

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If π has a descent at position i then $\pi_{i+1} \in I$.

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For example, if $\pi = 1432$ then certainly car 3 is a lucky car.

If π has a descent at position i then $\pi_{i+1} \in I$.

Theorem (Ferrerì, Harris, M., Swartz (2025))

Let $\pi \in \mathcal{O}_n(I)$ for some lucky set I .

Previous results

Theorem (Harris, M. (2024))

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Theorem (Ferreri, Harris, M., Swartz (2025))

Let $\pi \in \mathcal{O}_n(I)$ for some lucky set I .

Every permutation $\pi \in \mathcal{O}_n(I)$ admits a decomposition into $|I|$ increasing subsequences, where each subsequence begins with an element of the lucky set I .

Pattern avoiding parking permutations and lucky sets

Motivated by recent results on pattern avoidance and the lucky statistic for parking functions, we investigate the following question:

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Not all the pattern avoiding permutations arise as parking permutations with a fixed lucky set.

Example

Let $n = 5$ and $I = \{1, 2, 3\}$. Consider $\pi = 35412 \in \text{Av}_5(123)$. Since there is a descent at position $i = 2$, car 4 must belong to the set I . Hence, $\pi = 35412$ is not a 123-avoiding parking permutation.

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Given a fixed lucky set $I = [k]$, what structure is imposed on parking permutations with lucky set I that avoid a prescribed pattern?

Not all the pattern avoiding permutations arise as parking permutations with a fixed lucky set.

We focus on the case when $I = \{1, 2, \dots, k\} = [k]$.

Notation

Let $\Pi = \sigma_1, \sigma_2, \dots, \sigma_i$.

Let $\mathcal{O}_n(I; \Pi)$ denote the set of parking permutations with lucky set I that avoid each of the patterns in the set Π .

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Notice that $\mathcal{O}_n(I; \Pi) \subseteq \text{Av}_n(\Pi)$, with equality if $I = [n]$.

Pattern avoiding parking permutations and lucky sets

Definition

Fix a lucky set $I \subseteq [n]$. Then $\pi = \pi_1\pi_2 \cdots \pi_n \in \mathfrak{S}_n$ is a member of the set $\mathcal{O}_n(I; \sigma_1, \sigma_2, \dots, \sigma_i)$ if and only if

- 1 $\pi_1 \in I$,
- 2 if $\pi_{i-1} > \pi_i$, then $\pi_i \in I$, and
- 3 π avoids each of the patterns $\sigma_1, \sigma_2, \dots, \sigma_i$.

123-avoiding parking permutations

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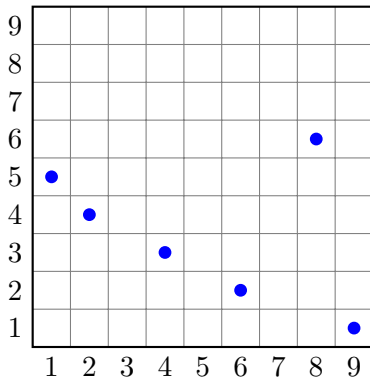
Example

Let $n = 5$ and $I = \{1, 2, 3\}$. The following are the only 123-avoiding parking permutations when the first 5 cars are lucky:

$$\mathcal{O}_5(I; 123) = \{25143, 32514, 35142, 35214, 35241\}.$$

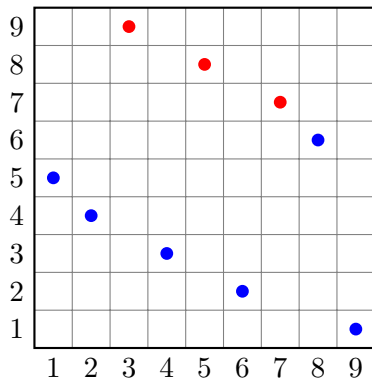
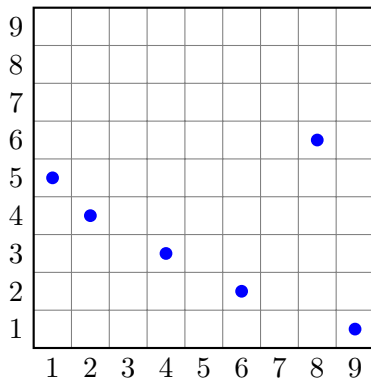
123-avoiding parking permutations (2)

Let $n = 9$ and $k = 6$ (the first six cars are lucky):



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Strategy:

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Strategy:

- (1) We will start with a $\sigma \in \text{Av}_k(123)$ formed by the lucky cars and
- (2) insert the $n - k$ unlucky cars in the permutation σ , where appropriate.

Structure of 123-avoiding parking permutations (2)

Start with:

$$\sigma_1 \quad | \sigma_2 \quad | \sigma_3 \quad \cdots \quad | \sigma_{k-1} \quad | \sigma_k$$

Structure of 123-avoiding parking permutations (2)

Need to insert $U = \{k + 1, k + 2, \dots, n\}$ after σ_1 .

$$\sigma_1 \underbrace{\nearrow} | \sigma_2 \underbrace{\nearrow} | \sigma_3 \underbrace{\nearrow} \cdots | \sigma_{k-1} \underbrace{\nearrow} | \sigma_k \underbrace{\nearrow}$$

Structure of 123-avoiding parking permutations (2)

Need to insert $U = \{k + 1, k + 2, \dots, n\}$ after σ_1 .

Let $\tau_i \in U$.

There are k possible positions to place $n - k$ elements:

$$\sigma_1 \tau_1 | \sigma_2 \quad | \sigma_3 \tau_2 \quad \cdots \quad | \sigma_{k-1} \quad | \sigma_k$$

Structure of 123-avoiding parking permutations (2)

Need to insert $U = \{k+1, k+2, \dots, n\}$ after σ_1 .

Since all elements in U are greater than any σ_i then whenever $\tau_i < \tau_j$ then the following cannot happen

$$\textcolor{blue}{\sigma_1} | \sigma_2 \textcolor{red}{\tau_i} \quad | \sigma_3 \tau_2 \quad \cdots \quad | \sigma_{k-1} \quad | \sigma_k \textcolor{red}{\tau_j}$$

as $\sigma_1 \tau_i \tau_j \equiv 123$.

Structure of 123-avoiding parking permutations (2)

Need to insert $U = \{k+1, k+2, \dots, n\}$ after σ_1 .

If $\tau_i < \tau_j$ then they must appear in decreasing order:

$$\sigma_1 | \sigma_2 \tau_j \quad | \sigma_3 \tau_2 \quad \cdots \quad | \sigma_{k-1} \quad | \sigma_k \tau_i$$

Structure of 123-avoiding parking permutations (2)

Need to insert $U = \{k+1, k+2, \dots, n\}$ after σ_1 .

Once we choose the positions for the elements in U , their order is fixed as they must appear in decreasing order from left to right. For $\tau_i < \tau_j$,

$$\sigma_1 | \sigma_2 \tau_j \quad | \sigma_3 \tau_2 \quad \cdots \quad | \sigma_{k-1} \quad | \sigma_k \tau_i$$

Question: In what positions can the unlucky cars be inserted?

123-avoiding parking permutations with lucky set $[k]$

Let j be the location of the first ascent of $\sigma \in \text{Av}_k(123)$, then

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If $A(k, j)$ be the number of 123-avoiding permutations (of length k) whose first ascent occurs at position j then

$$\sum_{i=1}^{n-k} \binom{i}{n-k} A(k, j)$$

enumerates the 123-avoiding parking permutations with lucky set $[k]$.

The location of the first ascent in $\text{Av}_n(123)$

Theorem (Connolly, Gabor, and Godbole (2015))

(k -fold Catalan convolution - OEIS [A033184](#)) The number of 123-avoiding permutations (of length n) whose first ascent is at position k is equal to the k -fold Catalan convolution:

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Hence, $A(k, j) = C_{k,j}$.

Enumeration of 123-avoiding parking permutations with lucky set $[k]$

Theorem (Ferreri, M. (2026))

(Lobb numbers - OEIS [A039599](#)) Let j be the location of the first ascent of $\sigma \in \text{Av}_k(123)$, then

$$\sum_{i=n-k}^{n-k} \binom{i}{n-k} \frac{j}{2k-j} \binom{2k-j}{k} = \frac{2n-2k+1}{n+1} \binom{2k}{n}$$

enumerates the 123-avoiding parking permutations with lucky set $[k]$.

What about other patterns?

Other patterns

- ① $\mathcal{O}_n([k]; 123)$ - Lobb numbers (OEIS [A039599](#)):

$$\frac{2n - 2k + 1}{n + 1} \binom{2k}{n}$$

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- ③ $\mathcal{O}_n([k]; 321)$ - new OEIS sequence:

$$\sum_{i=0}^{n-k} \binom{n-k}{i} \left(\frac{n-i-k+2}{n-i+1} \binom{n-i+k-1}{k-1} - \binom{n-i-2}{k-1} \right)$$

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- ❹ $\mathcal{O}_n([k]; 312)$ - new OEIS sequence:

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- ① Classify all combinations of two, three, four, five, or all six patterns of length three

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- ② General set of lucky cars
- ③ Connect this work with algebraic or geometric structures (quasisymmetric functions? polytopes?)

Acknowledgments

UC Denver and University of Denver for hosting me this week.

Thank You!

<https://marti310.github.io/research.html>

Structure:

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We construct them by first placing the unlucky cars and then placing the lucky cars.